Nahj AL-Balagha

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Overview

01 About NahjulBalagha

02 About LEX

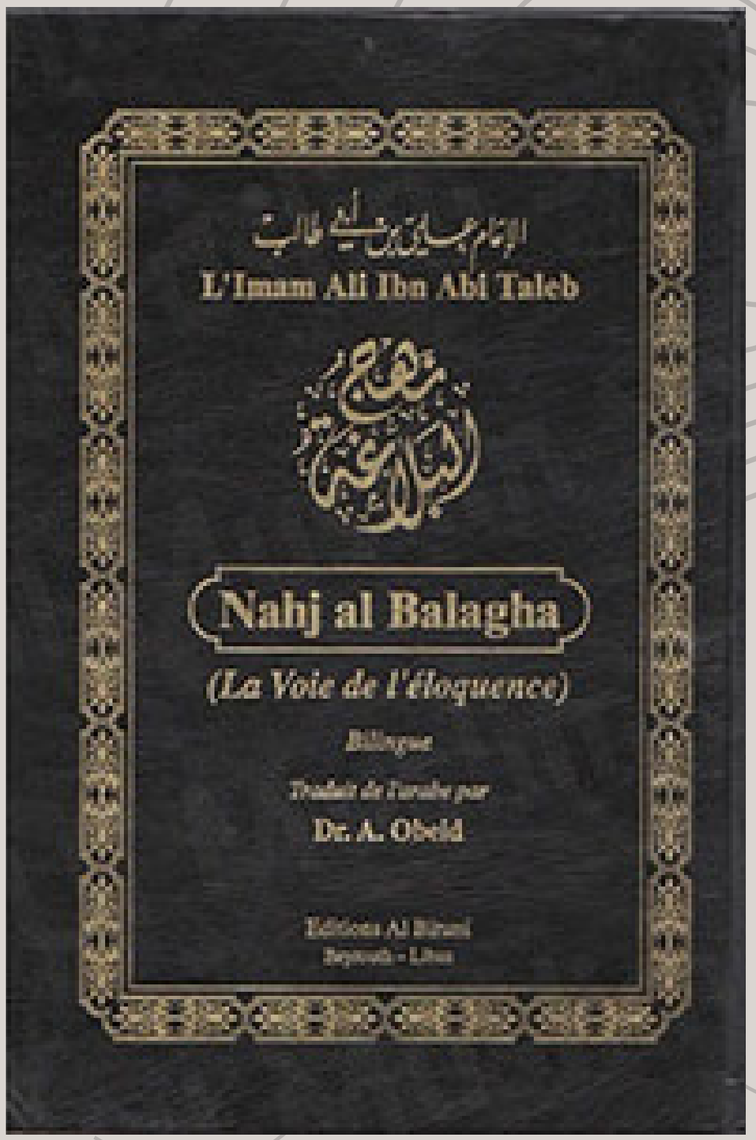
03 About YACC

Objective

Our mini-project revolves around employing modern compiler tools to extract and compile the profound wisdom encapsulated in select statements from Nahjul Balagha, a revered text in Islamic literature. By utilizing Lex for tokenization, Yacc for parsing, and semantic analysis techniques, our aim is to organize these statements into a structured format.

About Nahjul Balagha

A.K.A “The Path Of Eloquence” is a revered collection of sermons, letters, and sayings attributed to Ali ibn Abi Talib. Compiled by Sharif Razi in the 10th century, it's considered one of the most significant works in Islamic literature.
The number of sermons, letters, and statements ranged from 238 to 241, 77 to 79, and 463 to 489, respectively.



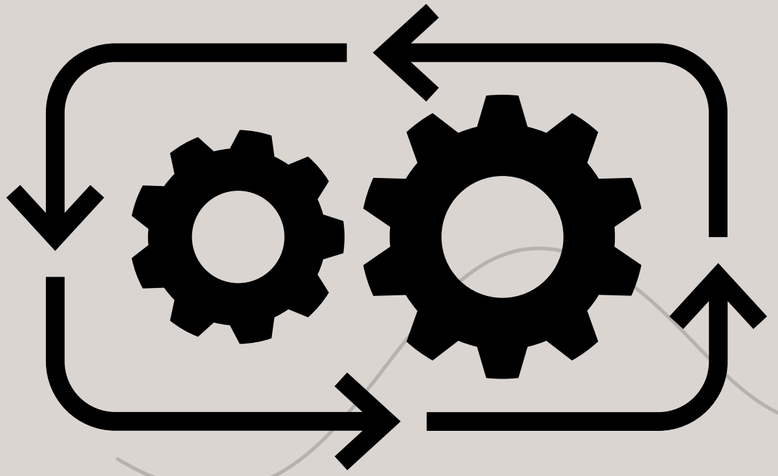
Compilation Process

1. Tokenization
2. Parsing
3. Semantic Analysis

About Source Code

We work on :

- 1326 Code Lines
- 25 Proverbs
- 227 Tokens
- 227 Expression Rules
- 25 Production Rules

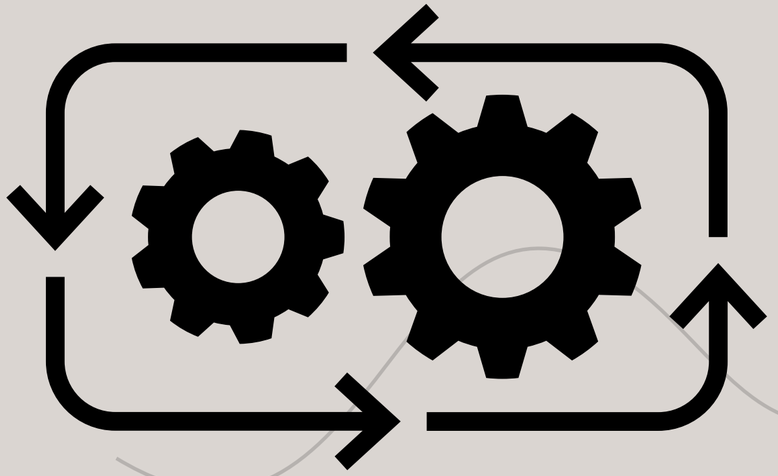


Lexical Analysis

Lex is primarily focused on constructing lexical analyzers or scanners for programming languages or text processing. It takes in a set of regular expressions and corresponding actions and generates code that can recognize patterns defined by those regular expressions in input text. Lexical analyzers built using Lex are responsible for tokenizing input, breaking it into meaningful units (tokens).

Parsing and Syntax Analysis

An acronym for "Yet Another Compiler Compiler," is a fundamental tool in computer science. It functions by processing a stream of tokens provided by a Lex file, where it not only parses these tokens but also conducts semantic analysis. Ultimately, Yacc generates source code in programming languages such as C, C++, or Java, enabling the construction of powerful compilers and interpreters.



Compilation Output

```
LexToken(ISSMMAWSOUL,'نم',1,0)
LexToken(FI3LMADI2,'عراص',1,3)
LexToken(MAF3OULBIH,'القَ',1,8)
LexToken(FI3LMADI,'هعرص',1,13)
(venv) PS C:\Users\Yassine\Desktop\ProjetCompilationFINAL>
```

Compilation Output

```
LexToken(ISSMMAWSOUL,'نم',1,0)
LexToken(FI3LMADI2,'صارع',1,3)
LexToken(MAF3OULBIH,'الحق',1,8)
LexToken(FI3LMADI,'صرعه',1,13)
BRAVO
```

Thank You !!

Overview

01 About NahjulBalagha

02 About Lex

03 Literary Preview

04 Theoretical

05 Objective

06 Hypothesis

07 Methodology

08 Implementation

Presented by Juliana Silva